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$y = m_1 x_1 + m_2 x_2 + b$

$\hat{y} = mx + b$

Iteration 2

$$m = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} \quad b = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix}$$

$$Xm = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} = \begin{pmatrix} -1.45 + 3\times(0.87) \\ \cancel{4\times(0.87)}\ 4\times(-1.45) + 10\times(0.87) \end{pmatrix} = \begin{pmatrix} -1.45+2.61 \\ -5.80+8.70 \end{pmatrix}$$

$$Xm = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix}$$

$$\hat{y} = Xm + b = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix} + \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} = \begin{pmatrix} 2.15 \\ 3.79 \end{pmatrix}$$

$$\hat{y} = \begin{pmatrix} 2.15 \\ 3.79 \end{pmatrix}$$

$$E = \hat{y} - y = \begin{pmatrix} 2.15 \\ 3.79 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$E = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

~~$MSE = 1/n (E^2) = 1/n ($~~

$$MSE = 1/n (E_1^2 + E_2^2) = 1/2 ((-2.85)^2 + (-2.21)^2)$$

$$MSE = \frac{8.1225 + 4.8841}{2} = \frac{13.0066}{2} = 6.5033$$

$$d\,MSE/dm = 2/n (X^T E) = (2/2)\left[\begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}\begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}\right] = \left[\begin{pmatrix} -2.85 + (4)(-2.21) \\ 3(-2.85 + (10)(-2.21) \end{pmatrix}\right]$$

$$d\,MSE/dm = \begin{pmatrix} -11.69 \\ -30.65 \end{pmatrix}$$

$$\frac{dMSE}{db} = \frac{2}{n} E = \frac{2}{2} \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$\frac{dMSE}{db} = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$m_{new} = m_{old} - \alpha \frac{dMSE}{dm} = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} - 0.01 \begin{pmatrix} -11.69 \\ -30.65 \end{pmatrix}$$

$$m_{new} = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} - \begin{pmatrix} -0.1169 \\ -0.3065 \end{pmatrix} = \begin{pmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{pmatrix}$$

$$m_{new} = \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix}$$

$$b_{new} = b_{old} - \alpha \frac{dMSE}{db} = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} - 0.01 \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$b_{new} = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} - \begin{pmatrix} -0.0285 \\ -0.0221 \end{pmatrix} = \begin{pmatrix} 0.99 + 0.0285 \\ 0.89 + 0.0221 \end{pmatrix}$$

$$b_{new} = \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix}$$